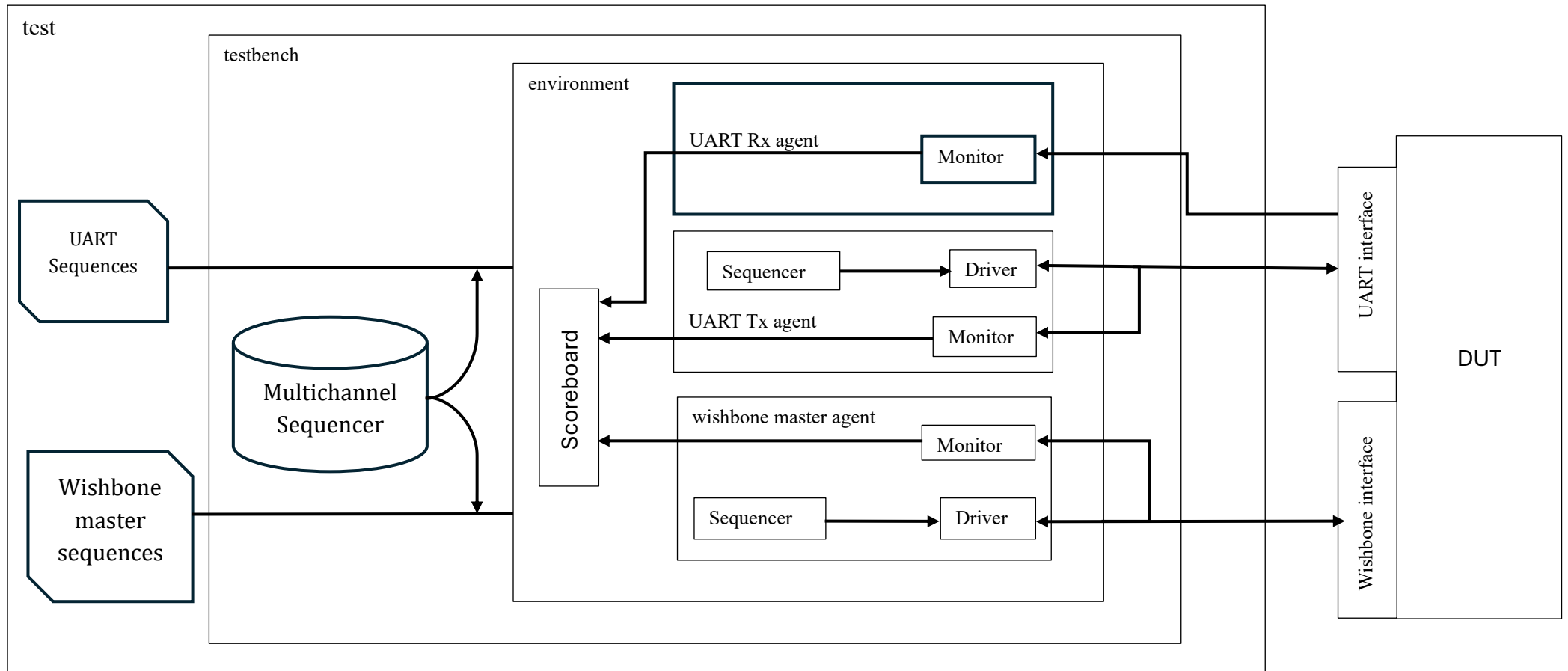


Testbench Architecture



Test Plan

No.	Test Name	Description	Test Type	Test Status	Comments
1.	Received Data available interrupt Test	This test will set the bit [0] of the Interrupt Enable Register IER. When this is set and data is available in receiver FIFO then the “Receiver Data available interrupt” should be asserted. At the same time bit [0] of LSR should also be 1. After this, reset the bit [0] of IER and then check for the “Receiver Data available interrupt”. It should not be asserted although the bit [0] of LSR is 1. If it still asserts then the test will be labelled as fail.	Directed		
2.	Transmitter Holding Register empty interrupt Test	This test will set the bit [1] of the Interrupt Enable Register IER. When this is set, and transmitter FIFO is empty then the “Transmitter Holding Register empty interrupt” should assert. At the same time, bit [5] or bit [6] of LSR should also be asserted. After this clear the bit [1] of IER, in this scenario the “Transmitter Holding Register empty interrupt” should not be asserted although the bit [5] of LSR is 1. If this is the scenario, then the test will be labelled as pass else fail.	Directed		
3.	Receiver Line Status Interrupt Test	This test will set the bit [2] of the Interrupt Enable Register IER. When this is set, “Receiver Line Status interrupt” should be asserted. At the same time, all or any of the bits from the bit [7, 4 : 1] of LSR is/are 1. After this, clear the bit [2] of IER, in this case the “Transmitter Holding Register empty interrupt” should not occur while the bit(s) [7, 4 : 1] of LSR will be 1. If this is the case then the test will be labelled as pass else fail.	Directed		
4.	Modem Status Interrupt Test	This test will set the bit [3] of the Interrupt Enable Register IER. When this is set, “Modem Status interrupt” should be asserted. In this scenario, any or all bit(s) of Modem Status Register in the range from 0-3 i.e. bit [3 : 0] can be 1. After this, disable the bit [3] of the IER. In this case the interrupt should not be asserted while the but(s) [3 : 0] are still 1. If this is the case then the test is labelled as pass else fail.	Directed		
5.	Reset test on Receiver and Transmitter FIFO	Write 1 to the bit [2 : 1] of the FIFO Control Register (FCR). If the FIFOs of transmitter and receiver reset their logic then the test will be labelled as pass else failed. This will be done by first writing some random data in these FIFOs and after this, these bits will be set. Also keep this in view that while this operation, the shift register of these FIFOs should not be affected they	Random		

		keep on performing their task else the test will be labelled as fail. If all scenarios satisfy, then the test will be labelled as pass.			
6.	Receiver FIFO Interrupt trigger level Test	This will be set by setting the different values of the bits [7 : 6] of the FCR and verify that when bits [7 : 6] = 00, Receiver FIFO interrupt should be triggered after 1 byte, in case of 01 it occurs after 4 bytes, in case of 10 it occurs after 8 bytes and in case of 11 this interrupt will occur after 14 bytes. If these conditions satisfy then the test will be labelled as pass else fail.	Directed		
7.	No. of bits in each character	Set different values of bits [1 : 0] of Line Control Register (LCR). When it is 00 there must be 5 bits in each character, in case of 01 there must be 6 bits, in case of 10 there should be 7 bits and in case of 11 there must be 8 bits in each character. After this try to transmit the data of these lengths and put the UART in loopback mode and observe the data received as well as the status of bit [0] of LSR. Bit[0] of LSR should be asserted after the character of certain length has been received. If it asserts before or after the settled character length then the test will be labelled as fail else pass.	Directed		
8.	No. of stop bits	Set the number of stop bits of data transmitted as 1 by clearing the bit[2] of LCR. Verify this by transmitting the data by keeping the UART in loopback mode and receive it in the same UART's receive FIFO and observe this. After this test, set this bit to 1 and observe that when this bit is asserted there must be 1.5 stop bits for the case when the configured character length is 5 bits and there are 2 stop bits in all other case. Also verify them by capturing the data stream inside the receiver FIFOs by configuring the UART in loopback mode. If these conditions satisfy, then the test will be labelled as pass else fail.	Random		
9.	Parity Enable and Even/Odd Test	Set the bit [3] of LCR to enable the parity check on each incoming and outgoing character. After this set the bit [4] of LCR to 1 so that even parity is transmitted and received in each character. After this test clear the bit [4] of LCR so that odd parity is generated and received in each transaction. If the UART's transmission follows this behavior then label the test as pass else fail	Directed		
10.	Stick parity Test	If bit [5] of LCR is 0 then no stick parity bit is checked. If it is set and bit [3] and bit [4] are 1 then parity bit is transmitted and received as 0 and if bit [5] is 1 as well as bit [3] is 1 and bit [4] is 0 then parity it is transmitted and	Directed		

		received as 1. Verify this behaviour and if the DUT performs accordingly then label the test as pass else fail.			
11.	Break control Test	If bit [6] of LCR is 1 then the serial out is forced to drive 0 at the output instead of some other meaningful data and if its 0 then normal transmission of data will take place. If DUT behaves accordingly then label the test as pass else fail.	Directed		
12.	Divisor latch access bit Test	If bit [7] of LCR is set then only in that case the divisor latches will be accessed. They are used to set the baud rate for the UART's communication without which the communication cannot be started. Try to set the baud rate when this bit is 0, if baud rate will be set and UART starts communication then there's an error regarding it's functionality.	Directed		
13.	Modem Test	Test this by clearing the bit [0] and bit [1] of Modem Control register and the values of CTS, DCD, RI and DSR are provided from sequencer, and this protocol will be verified.	Random		

Coverage Plan

No	Covergroup Name	Description	Coverage Achieved/Not
1	data_i cover	All possible 8 bit as well as 32 bit values of data_i should be covered	
2	addr_i_cover	All possible values of input address should be covered	
3	sel_i_cover	All possible values of sel_i should be covered.	